FH Vorarlberg
University of Applied Sciences

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Course Number 074742056202

Energy Storage

Applications for Electrical Energy Storage Systems (EESS)

- Power Quality Improvements
  - Prevent voltage sags, flickers
- Uninterrupted Power Supply
  - Bridge the time until backup power is in service
- Energy Management
  - Peak shaving, power balancing
  - Energy refinement

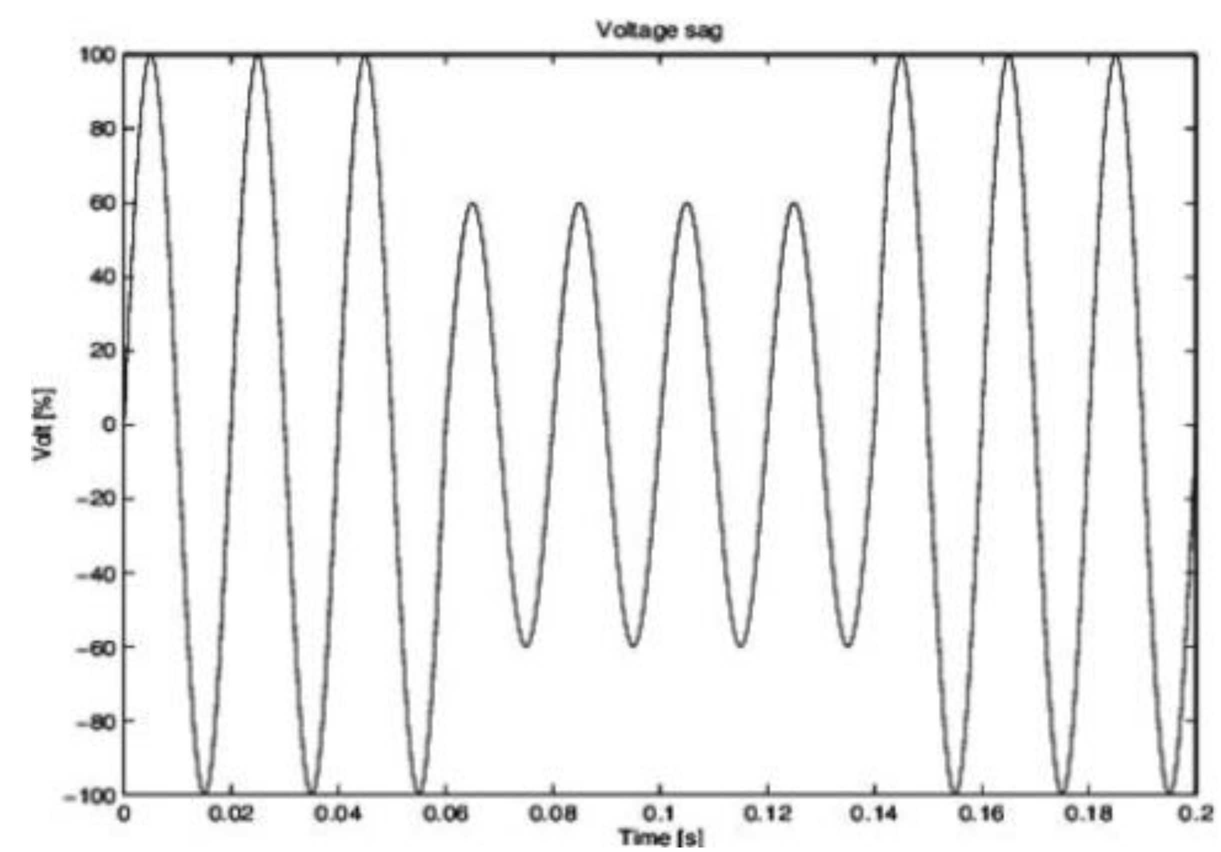


Figure: 40% voltage sag with a duration of 80 ms
Source: researchgate.net

Power Quality Improvements

- Time range: within seconds, up to minutes
- Technologies
  - Fly wheels
  - Magnetic energy storage
  - Super capacitors
  - Different types of batteries

Uninterrupted Power Supply

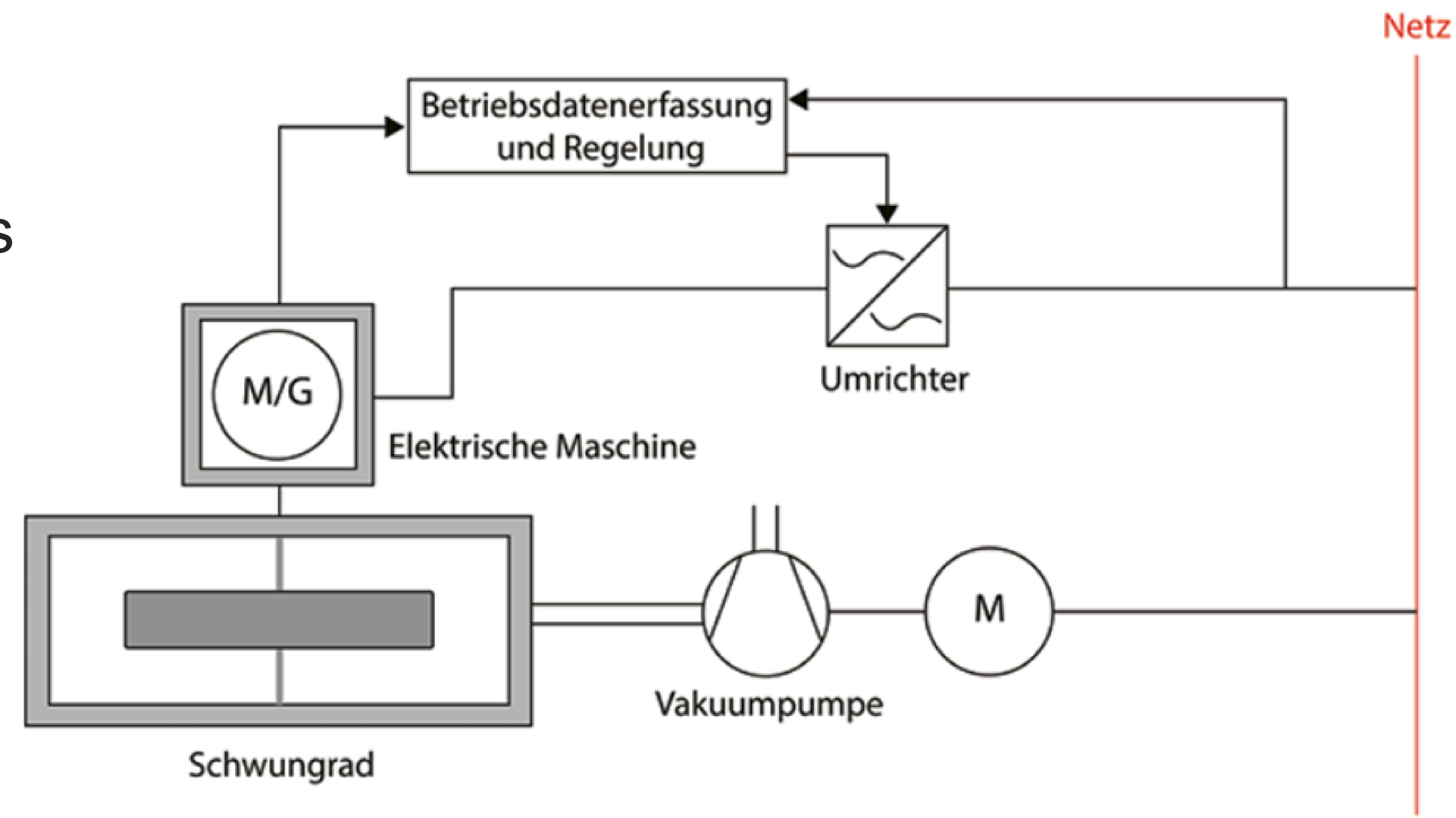
- Time range: within minutes
- Technologies
  - Super capacitors
  - Different types of batteries

Energy Management

- Time range: 15 min and more
- Technologies
  - Pump storage power plants
  - Compressed Air energy Storage (CAES)
  - High energy batteries
  - Power to gas

Fly Wheels - History

- Rotating pottery disk
- Steam machines - Balancing rotary motion
- Combustion engines – piston forces to mechanical transmission
- 1883: John A. Howel
  - 160 kg steel disk
  - 21.000 rev/min
  - acc. torpedo 55 km/h, 1.5 km through water



Fly Wheel - History

- 1950 – Gyrobus
  - Swiss company Oerlikon
  - Stored 9.15 kWh
  - Possible to charge
    - during breaking
    - at bus stops
  - Fly wheel made of steel
  - turning in hydrogen atmosphere
  - possible to travel 3.5 km

Fly Wheel - History

- 1954 – 1966 – Gyro-Locomotivew
  - One for a mine in Lothringen
  - Three for mines in Johannesburg, South Africa
- 1973 – Concept car by Richard F. Post.
  - fiber-composite material
  - Charge 30 kWh
  - 320 km with 100 km/h
- Several other, single application have been build in the years after

Fly Wheel - Physics

- The inertia *J* depends on the mass *m* and the radius *r*

- The storable energy E_{kin} depends on inertia and rotational speed omega

$$J = mr^2$$

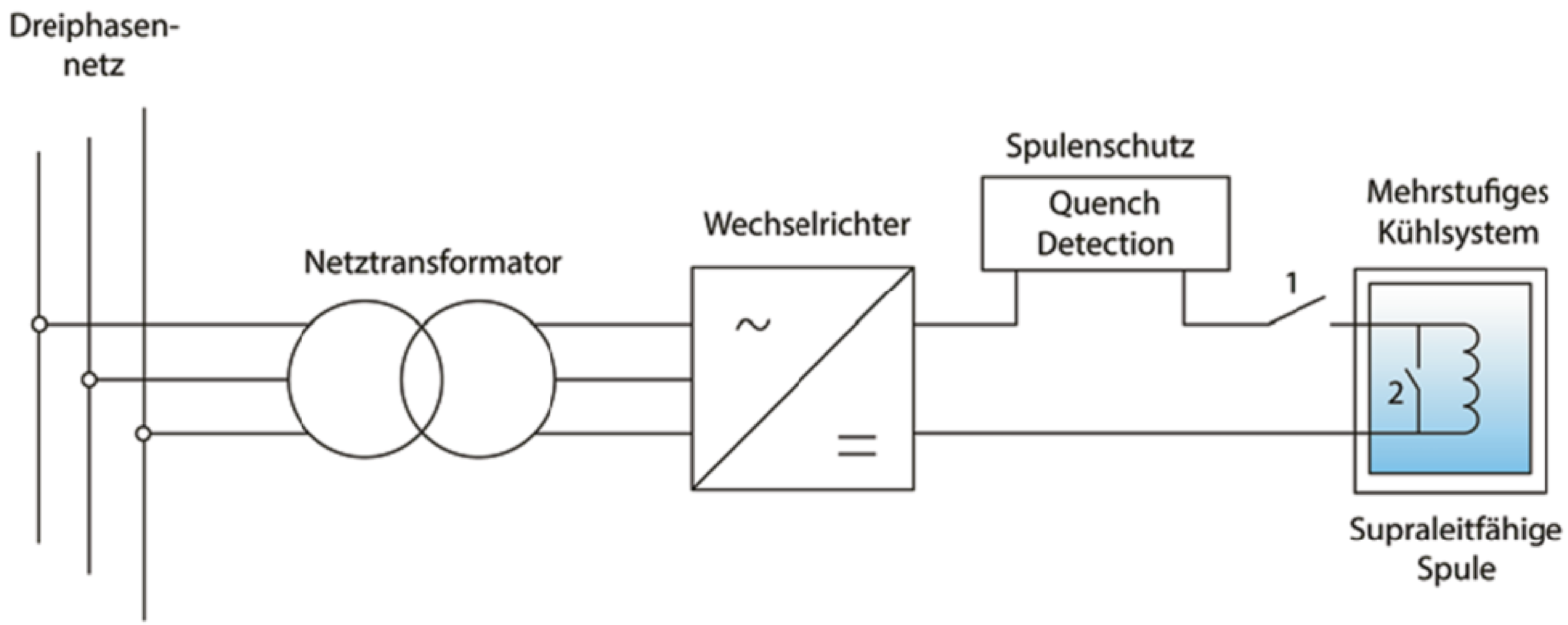
$$E_{kin} = \frac{1}{2}m(r\omega)^2$$

Fly wheels – Technical challenge

- Challenge – getting the right material
- Prone to collapse during lifetime

- Picture left: destroyed flying wheel made of composite material
- Picture right: flying wheel made of steel


Magnetic Energy Storage



- Superconducting electro-magnetic energy storage (SMES)
  - superconducting coil
  - converter
  - protection components (quench detector)

Magnetic Energy Storage

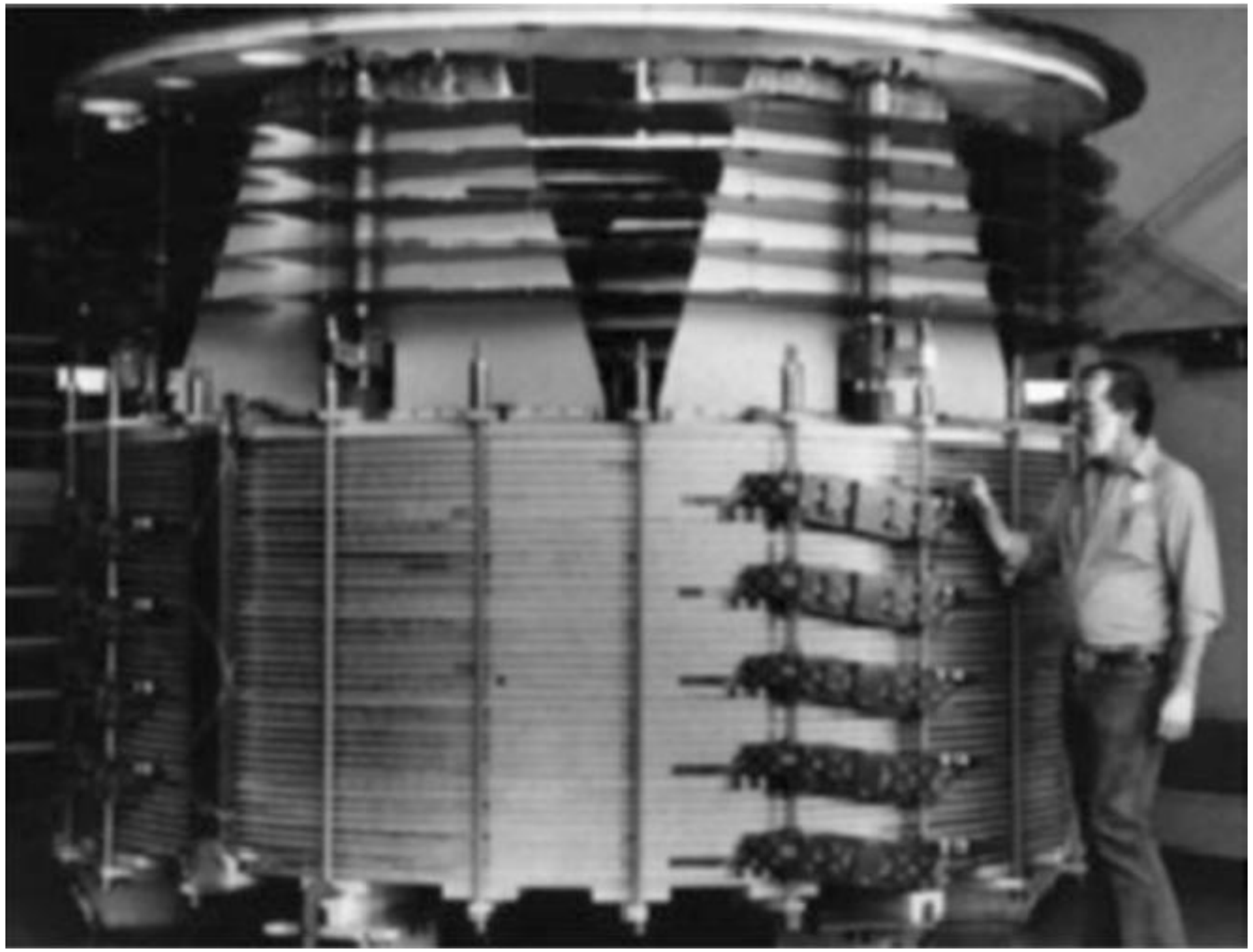
- Coil is cooled down by cryogenic elements
  - helium, hydrogen, nitrogen, argon, oxygen, dry ice

- Apply DC to Coil -> current starts to flow -> Magnetic Field
- Switch off supply –> Current flows out of the coil
- During discharge: coil acts like voltage source
- Superconducting state eliminates ohmic resistance

Applications

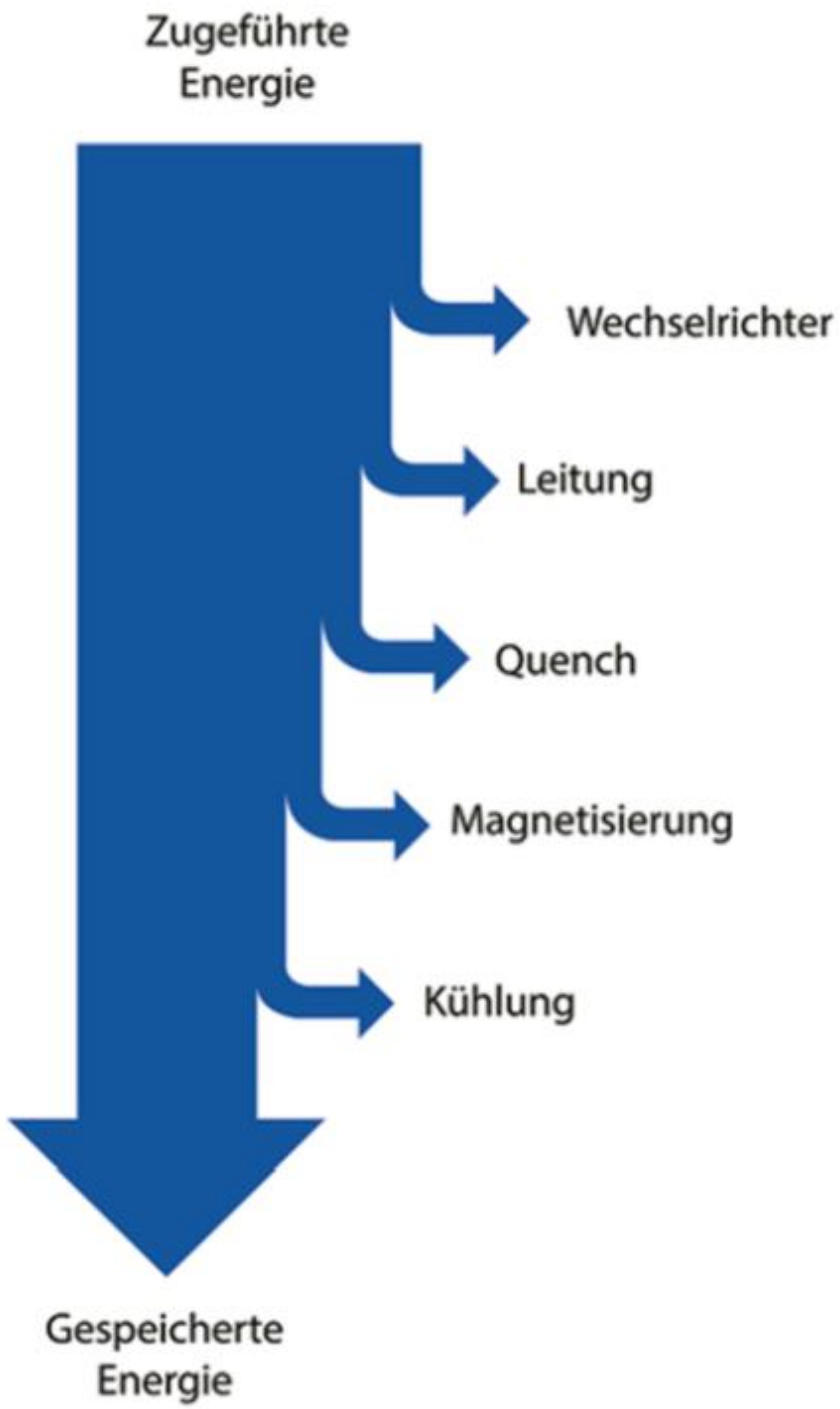
- Only research applications
- 1 Million charge/discharge cycles
- 30 years expected life time

- Drawback energy losses



30 MJ coil at Los National Laboratoy

Energy Losses



- Losses by inverter, cables, protection device, magnetizing, cooling
- Cooling is very expensive
  - -269 °C; Helium is needed (expensive)
  - -196 °C; Nitrogen could be used (cheaper)
- Study of NEDO (Japanese research)
  - Power plant 110 MW: 1 kW costs 690 Dollar
  - Power plant 100 MW: 1 kW costs 1970 Dollar

Super Capacitors

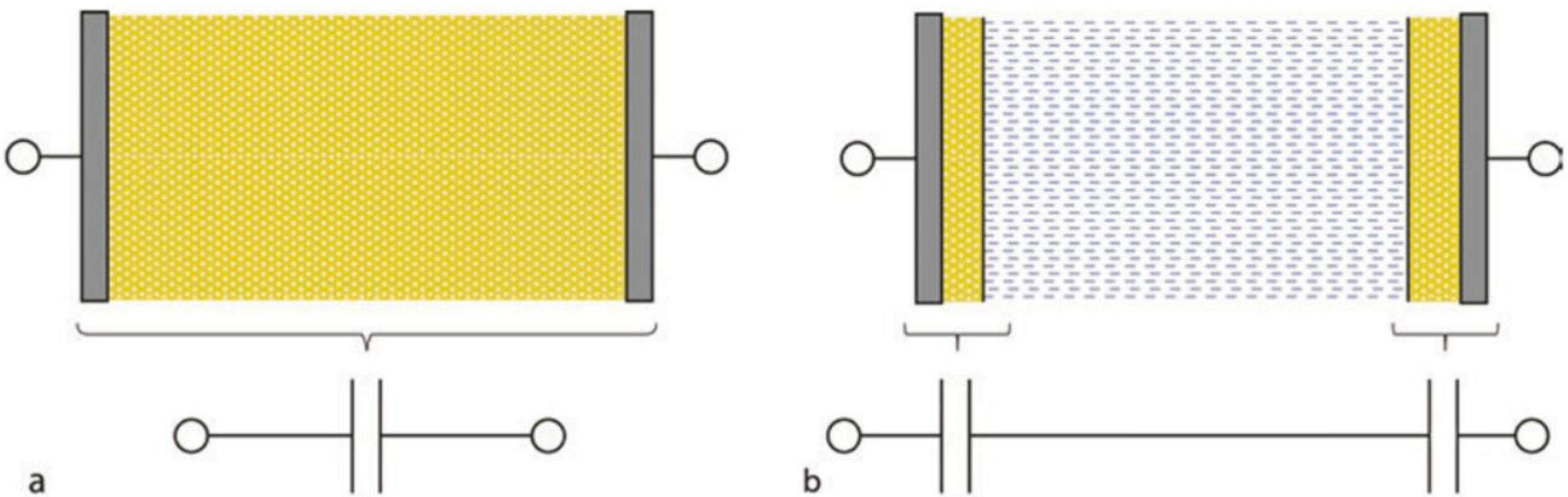


Abb. 6.5 **a** Der »klassische« Kondensator, Elektrode – Dielektrikum – Elektrode; **b** der Superkondensator, Elektrode – dielektrische Schicht – Elektrolyt – dielektrische Schicht – Elektrode

Supercaps are designed multilayered or in coil form

Super caps

- Energy density
  - 65 Farad capacitors - 3.9 Wh/kg
  - 5 Farad capacitors - 2.2 Wh/kg
- Lifetime expectation
  - usually 10 years
  - industrial graded: 15 years
- Charging cylces
  - 500.000 to 1.000.000

Super cap

- Capacitors are self-discharging
  - irregularities in the distance between charge carriers
  - results in a low exchange of load carriers and gradual discharge (aka leakage current)
- Influenced by
  - Capacitance
  - Voltage
  - Temperature
  - chemical stability of electrode-electrolyte combination
- Example
  - Panasonice Goldcapacitor 5.5 V/F
  - Voltage drop at 20 °C from 5.5 V to 3 V in 600 h

Application

- Supercap Application in a hub of a Wind turbine

Batteries

- Lead Acid cells
- Nickel batteries
- Lithium batteries
- Sodium sulfur (dt. Natrium Schwefel) batteries

Lead Acid Cells

- Operating at room temperature
- Energy density: 120 Ah/kg
- Main application
  - Starter battery
  - Electric drive battery
  - Emergency power supply

Nickel Batteries

- Goal of development
  - Higher energy density
  - Better reliability
- Lower cell voltage: 1.4 V (lead acid 2.1 V)
- Higher energy density: 224 Ah/kg (lead acid 120 Ah/kg)
- Electrolyte is less prone to aging, works on lower temperature

Lithium batteries

- High temperature batteries
- High cell voltage: 3.5 V
- Energy density: 286 Ah/kg
- Need battery management system
  - control charging / discharging cycle
  - control cell temperature and voltage

Sodium Sulfur

- Cell voltage: 2.1 V
- Energy density: 377 Ah/kg
- High temperature: 300 °C

- Possible to scale up to a wind farm requirements (35 MW)

Figure: 1 MQ NaS Battery from Younicos AG Berlin

Compressed Air Energy Storage

- Two commercial operated plants worldwide (Huntdorf, Germany; USA)
- Huntdorf, build 1978
- Two caverns
  - salt mine
  - 650 to 800 m depth
  - Volume 310.000 m³
  - Combined gas turbine, pressure turbine

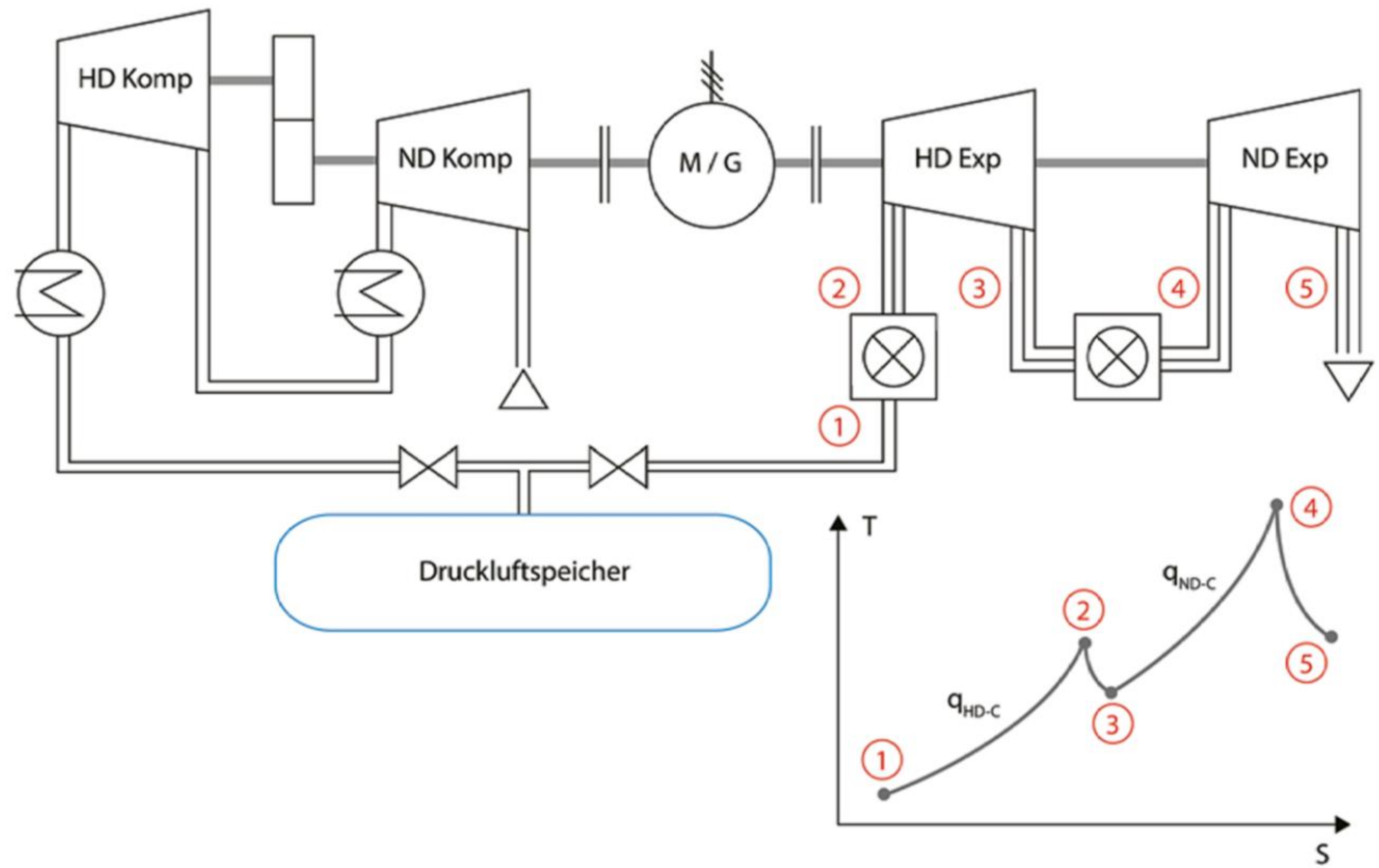


Figure: Power Plant Huntorf setup

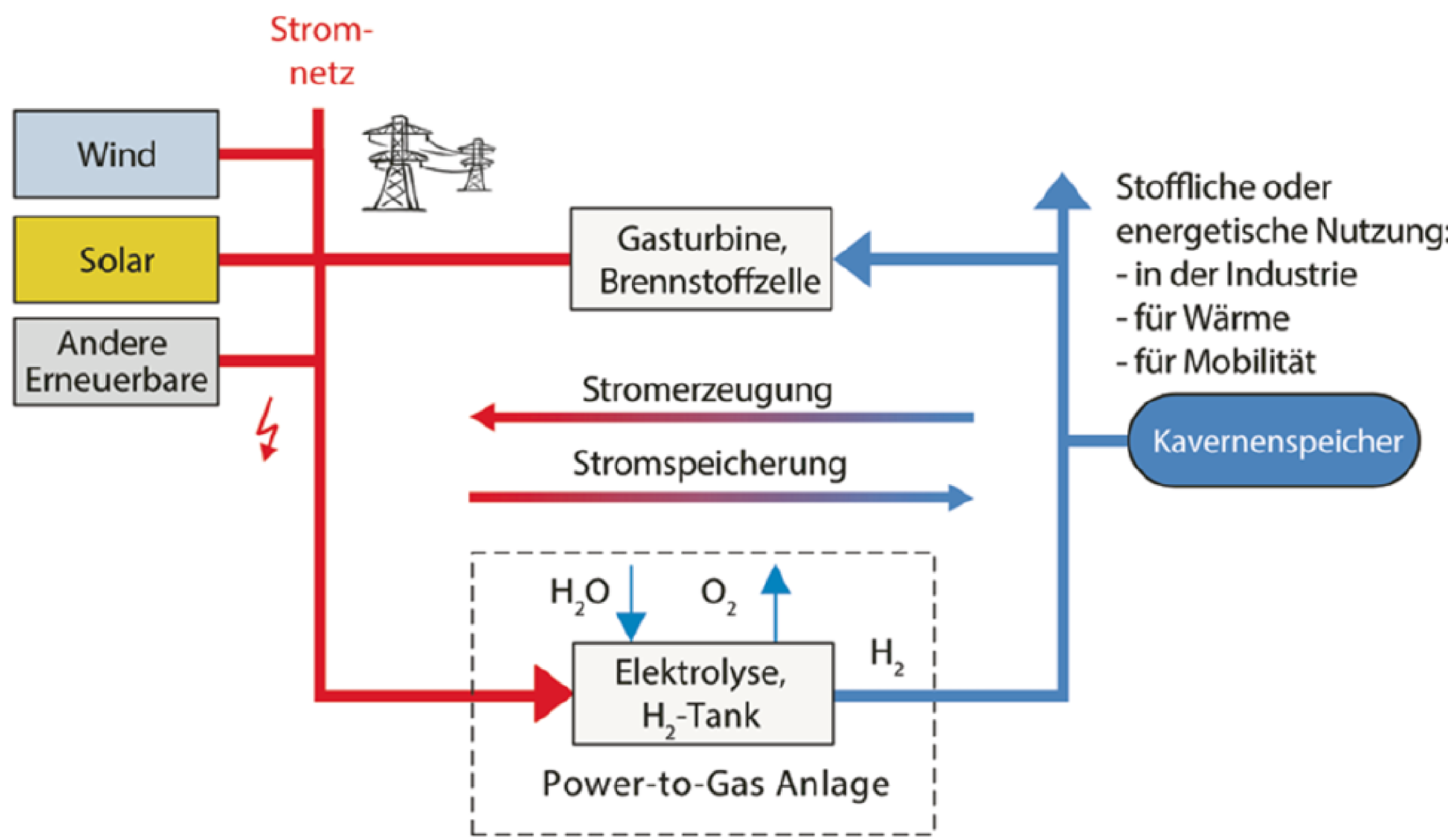
Compressed Air

- Huntorf
  - Air is pressed into cavern
  - up to 72 bar
  - Compressor 60 MW, for 2h
  - Air has to be
    - cooled during compression
    - heated during taking it out of the cavern
  - Electrical energy rating: 321 MW

Power To Gas

- Electricity from fluctuating energy producers
- Create hydrogen (Electrolyze)
- Hydrogen read for transportation (80 bar)

- Overall efficiency
  - 50 to 80 %



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